TECHNICAL HANDOVER SPECIFICATION

BeatWave PRO Design System

Tokens, components, accessibility rules and frontend architecture for the CleanDJmix / BeatWave PRO web client.

01 — Brand & Philosophy

02 — Color System

03 — Typography & Spacing

04 — Component Library

05 — Icons & Containers

06 — Accessibility Matrix

07 — Architecture

DOCUMENT	VERSION	REVISION
ANS-BW-001	v1.0.0	2026-11-06

Precision audio intelligence, designed for the dark booth.

This document is the single source of truth for the BeatWave PRO interface: tokens, components, accessibility rules and the frontend architecture. Every value here maps 1:1 to a CSS variable or Tailwind class in the codebase.

PRODUCT	TARGET STACK	DOCUMENT	REVISION
BeatWave PRO (CleanDJmix)	Next.js App Router · React · Tailwind v4 · TS	ANS-BW-001 · v1.0.0	2026-11-06



FIG. 1 Production Cue Editor. Note how saturated colour is reserved for the waveform, the active pad and the primary actions — the rest of the chrome stays neutral.

DESIGN PRINCIPLES

True low-light

Dark mode is the primary environment. Canvas is #09090B, never pure black, so surfaces can step up in luminance without halos.

DON'T · Do not use white fills larger than a button in dark mode.

Selective exposure

Only the active deck, playhead and the focused control carry saturated color. Everything else stays in the zinc ramp.

DON'T · Do not colour idle pads, inactive tabs or static labels.

Data before decoration

BPM, key, time and dB are the product. They render in tabular mono so values never jitter while a track plays.

DON'T · Do not animate numbers or add icons that are not actionable.

One codebase, every booth

Desktop booth, laptop and tablet remote share components. Layout adapts; controls keep a 44px minimum touch target.

DON'T · Do not hide core transport controls behind menus on small screens.

VOICE & TONE

Short, technical, booth-ready. Labels read like hardware silkscreen — `CUE`, `SYNC`, `SNAP: ON` — and help text explains outcomes, not features.

WORDMARK



Always on the dark canvas. Minimum clear space equals the cap height of “B”.

§ 02 / 07

Color system

Eleven semantic tokens drive every surface. Components only ever reference semantic tokens; the static ramps exist for data-viz and brand artwork where a value must not flip with the theme.

SEMANTIC TOKENS

TOKEN	LIGHT	DARK	TAILWIND	ROLE
--background	#FAFAFA L · rgb(250 250 250)	#09090B D · rgb(9 9 11)	bg-background	App canvas, page body
--surface	#FFFFFF L · rgb(255 255 255)	#121216 D · rgb(18 18 22)	bg-surface	Cards, decks, panels
--surface-2	#F4F4F5 L · rgb(244 244 245)	#18181F D · rgb(24 24 31)	bg-surface-2	Nested wells, inputs, code
--border	#E4E4E7 L · rgb(228 228 231)	#2A2A35 D · rgb(42 42 53)	border-border	Hairlines, dividers, outlines
--foreground	#09090B L · rgb(9 9 11)	#F4F4F5 D · rgb(244 244 245)	text-foreground	Primary text, icons
--muted-foreground	#52525B L · rgb(82 82 91)	#A1A1AA D · rgb(161 161 170)	text-muted-foreground	Secondary text, labels
--primary	#0E7490 L · rgb(14 116 144)	#22D3EE D · rgb(34 211 238)	bg-primary / text-primary	Brand cyan — active, BPM, focus
--primary-foreground	#FFFFFF L · rgb(255 255 255)	#09090B D · rgb(9 9 11)	text-primary-foreground	Text on primary fills
--success	#15803D L · rgb(21 128 61)	#4ADE80 D · rgb(74 222 128)	bg-success / text-success	Analyzed, saved, mix-match

TOKEN	LIGHT	DARK	TAILWIND	ROLE
--warning	#B45309 L · rgb(180 83 9)	#FBBF24 D · rgb(251 191 36)	bg-warning / text-warning	CUE, pending, attention

STATIC RAMPS

DJ Zinc (Neutral)

bg-djzinc-{step}

950	925	900	850	800	750	700	600
#09090B	#101014	#121216	#18181F	#202028	#2A2A35	#32323F	#484858

DJ Cyan (Primary)

bg-djcyan-{step}

200	300	400	500	600	700	800	900
#A5F3FC	#67E8F9	#22D3EE	#06B6D4	#0891B2	#0E7490	#155E75	#164E63

WAVEFORM SIGNAL BANDS

Frequency colours are fixed in both themes so DJs can read the waveform the same way as on Serato / Rekordbox hardware. Never use them for UI state.

Sub / Bass	< 150 Hz	#3B82F6
Low-Mid / Drums	150–800 Hz	#EF4444
Mid-High / Vocals	800 Hz–4 kHz	#F59E0B
Treble / Hats	> 4 kHz	#22D3EE

TAILWIND MAPPING

GLOBALS.CSS

```
/* app/globals.css — Tailwind v4 (no tailwind.config.js) */
:root  { --primary: #0e7490; --background: #fafafa; /* … */ }
.dark  { --primary: #22d3ee; --background: #09090b; /* … */ }

@theme inline {
  --color-background: var(--background);
  --color-surface:    var(--surface);
  --color-primary:    var(--primary);
  --color-success:    var(--success);
  --color-warning:    var(--warning);
  --color-destructive:var(--destructive);
  /* static ramps → bg-djzinc-900, text-djcyan-400 */
  --color-djzinc-900: #121216;
  --color-djcyan-400: #22d3ee;
}
```

§ 03 / 07

Typography & spacing

Two families, a 1.2 modular type scale and a 4px base grid. Body text never drops below 14px; mono labels at 12px are reserved for hardware-style meta, never paragraphs.

Inter `font-sans`

Cobblestone Jazz — W

UI, headings, body copy

JetBrains Mono `font-mono`

128.0 BPM · 8A · 00:01.50

BPM, key, time, dB, labels, code

MODULAR TYPE SCALE

Display 48px · 3rem · lh 1	Peak time …	`text-5xl font-semibold tracking-tight`	Hero / document title
H1 36px · 2.25rem · lh 1.11	Peak time selec…	`text-4xl font-semibold tracking-tight`	Page titles
H2 30px · 1.875rem · lh 1.2	Peak time selection	`text-3xl font-semibold tracking-tight`	Section headers
H3 24px · 1.5rem · lh 1.33	Peak time selection	`text-2xl font-semibold`	Panel titles
H4 20px · 1.25rem · lh 1.4	Peak time selection	`text-xl font-medium`	Card titles, track names

Body L 18px · 1.125rem · lh 1.625	Peak time selection	`text-lg leading-relaxed` Lead paragraphs
Body 16px · 1rem · lh 1.625	Peak time selection	`text-base leading-relaxed` Default UI copy
Small 14px · 0.875rem · lh 1.5	Peak time selection	`text-sm leading-6` Dense tables, inputs

SPACING GRID (4PX BASE)

0.5	2px / 0.125rem	Hairline offsets
1	4px / 0.25rem	Icon ↔ text in badges
2	8px / 0.5rem	Inline gaps, button icon gap
3	12px / 0.75rem	Compact control padding
4	16px / 1rem	Default component padding
6	24px / 1.5rem	Card padding, grid gaps
8	32px / 2rem	Panel padding
12	48px / 3rem	Section rhythm (mobile)
16	64px / 4rem	Section rhythm (desktop)
24	96px / 6rem	Page-level separation

Spacing is always applied with `gap-*` on flex/grid parents. Never mix margin/padding with gap on the same element, and never use `space-*`.

CORNER RADIUS

`rounded-sm`
2px

`rounded-md`
4px

`rounded-lg`

`rounded-xl`

6px

10px

Base `--radius: 0.375rem`. Circular controls (play, CUE, knobs) use `rounded-full`.

§ 04 / 07

Component library

Every interactive primitive ships with five documented states. Use the state switcher to force a state and read the exact class binding; choose Live to interact with the real component.

Button / Primary

One per view. Commits a destructive-to-disk or high-value action (save track, export batch).



CLASSNAME · DEFAULT

```
inline-flex h-10 items-center gap-2 rounded-md bg-primary px-4 text-sm font-semibold text-primary-foreground transition-colors hover:bg-primary/90 active:bg-primary/80 disabled:pointer-events-none disabled:opacity-40 focus-visible:outline-none focus-visible:ring-2 focus-visible:ring-ring focus-visible:ring-offset-2 focus-visible:ring-offset-background
```

Button / Secondary

Neutral outline button for navigation and reversible actions (change folder, view list).

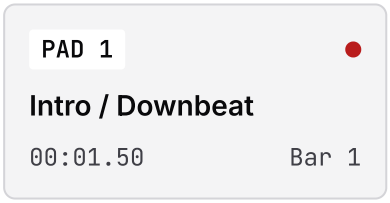
View list (12)

CLASSNAME · DEFAULT

```
inline-flex h-10 items-center gap-2 rounded-md border border-border bg-surface px-4 text-sm font-medium text-foreground transition-colors hover:bg-surface-2 active:bg-border disabled:pointer-events-none disabled:opacity-40 focus-visible:outline-none focus-visible:ring-2 focus-visible:ring-ring focus-visible:ring-offset-2 focus-visible:ring-offset-background
```

Hot Cue Pad

Performance pad. Idle pads stay neutral; colour appears only when the pad is set or triggered.



CLASSNAME · DEFAULT

```
flex w-48 flex-col gap-2 rounded-lg border border-border bg-surface-2 p-3 text-left transition-colors hover:border-destructive/60 active:bg-destructive/15 disabled:pointer-events-none disabled:opacity-40 focus-visible:outline-none focus-visible:ring-2 focus-visible:ring-ring focus-visible:ring-offset-2 focus-visible:ring-offset-background
```

Transport / Play

56px circular transport control. Always visible, always keyboard-reachable (Space).



CLASSNAME · DEFAULT

```
inline-flex size-14 items-center justify-center rounded-full border-2 border-success text-success transition-colors hover:bg-success/10 active:bg-success active:text-background disabled:pointer-events-none disabled:opacity-40 focus-visible:outline-none focus-visible:ring-2 focus-visible:ring-ring focus-visible:ring-offset-2 focus-visible:ring-offset-background
```

Toggle Chip

Binary mode switch for the waveform toolbar (Snap, Grid, Follow). Uses aria-pressed.



CLASSNAME · DEFAULT

```
inline-flex h-8 items-center gap-1.5 rounded-md border border-border bg-surface px-3 font-mono text-xs uppercase tracking-widest text-muted-foreground transition-colors hover:text-foreground aria-pressed:border-primary/60 aria-pressed:text-primary disabled:pointer-events-none disabled:opacity-40 focus-visible:outline-none focus-visible:ring-2 focus-visible:ring-ring focus-visible:ring-offset-2 focus-visible:ring-offset-background
```

Search Input

Library filter. Visible label for screen readers, 16px text to avoid iOS zoom.

Artist, title, key...

CLASSNAME · DEFAULT

```
h-10 w-64 rounded-md border border-border bg-surface-2 pl-9 pr-3 text-base text-foreground placeholder:text-muted-foreground transition-colors hover:border-muted-foreground/50 disabled:cursor-not-allowed disabled:opacity-40 focus-visible:outline-none focus-visible:ring-2 focus-visible:ring-ring focus-visible:ring-offset-2 focus-visible:ring-offset-background
```

Status Badge

Non-interactive. Tinted fill at 10%, border at 40%, text at full token. Never rely on colour alone — the label carries meaning.

128.0 BPM 8A Analyzed Pending Clipping

```
rounded-sm border px-2 py-0.5 font-mono text-xs font-semibold border-primary/40 bg-primary/10 text-primary
```

COMPONENTS IN PRODUCTION



FIG. 2 Waveform & cue grid. Toolbar uses Toggle Chips; markers use the signal-band palette.

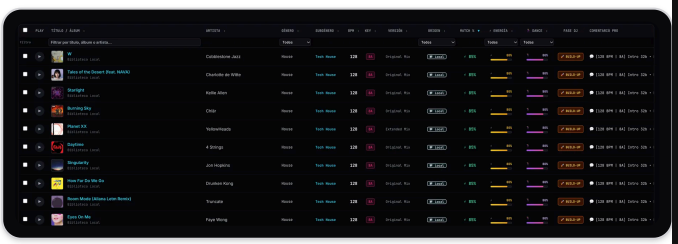


FIG. 3 Library table. Dense Small (14px) rows, mono numerics, Status Badges per row.

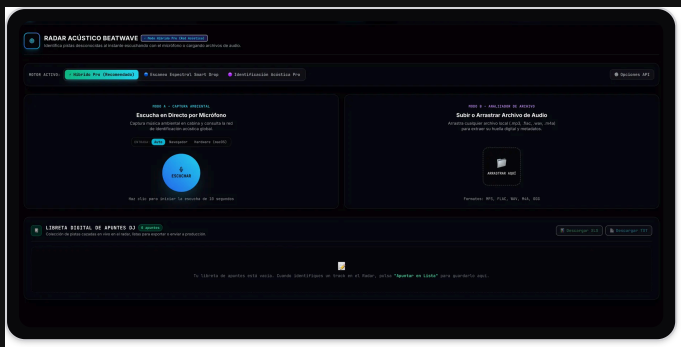


FIG. 4 Recognition panel. Surface card on canvas with a single primary action.

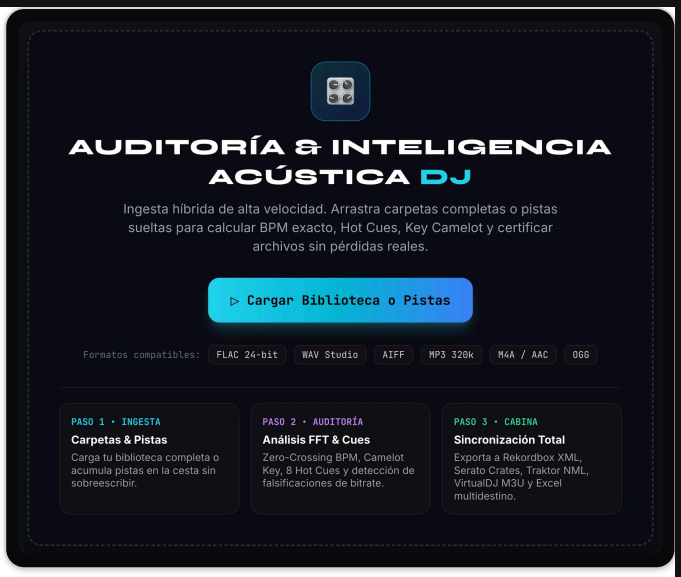


FIG. 5 Drop zone. Dashed border-border at rest, border-primary on drag-over.

§ 05 / 07

Icons & containers

Lucide React is the only icon set. Stroke width stays at the default 2px, icons inherit currentColor, and three sizes cover every case. Layout is mobile-first across six breakpoints.

STANDARD ICON SET · LUCIDE-REACT

Play	Pause	SkipBack	Disc3	AudioWaveform
Headphones	Volume2	SlidersHorizontal	Magnet	ZoomIn
Library	ListMusic	FolderOpen	Search	Radar
Sparkles	Save	Download	Trash2	Settings

SIZES

`size-4 · 16px`
Inline with text, badges, dense buttons

`size-5 · 20px`
Default button & nav icons

`size-6 · 24px`
Transport controls, empty states

USAGE

TSX

```tsx
import { Play } from 'lucide-react'

// Decorative (label is visible text): hide from AT
<button className="inline-flex items-center gap-2">
  <Play className="size-4" aria-hidden="true" /> Play
</button>

// Icon-only: the button carries the accessible name
<button aria-label="Play deck 1">
  <Play className="size-6" aria-hidden="true" />
</button>
```

BREAKPOINTS & CONTAINERS

PREFIX	MIN WIDTH	RANGE	CONTAINER	LAYOUT
base	0px		100% · px-4	Single column, bottom tab bar (Library · Acoustic · Cue Editor)
sm	640px		100% · px-6	Single column, wider controls
md	768px		100% · px-6	Two columns: library + inspector drawer
lg	1024px		max-w-7xl · px-8	Three-column dashboard, side panels visible
xl	1280px		max-w-7xl · px-8	Full deck + waveform + hot-cue grid
2xl	1536px		max-w-screen-2xl · px-8	Club-booth widescreen, dual deck

DASHBOARD-SHELL.TSX

```
<div className="mx-auto w-full max-w-7xl px-4 sm:px-6 lg:px-8">
  <div className="grid gap-6 md:grid-cols-[18rem_1fr] xl:grid-cols-
[18rem_1fr_20rem]">
    <LibraryPanel />  <DeckWorkspace />  <InspectorPanel className="hidden xl:flex"
/>
  </div>
</div>
```

§ 06 / 07

Accessibility matrix

WCAG 2.1 AA is the release gate; AAA is the target for body text. Contrast ratios below are computed live from the token values, so the matrix can never drift from the palette.

CONTRAST RATIOS (COMPUTED)

PAIR	LIGHT	DARK
`--foreground` / `--background` Body text on canvas	Aa 19.06:1 ✓ AA-L ✓ AA ✓ AAA	Aa 18.10:1 ✓ AA-L ✓ AA ✓ AAA
`--foreground` / `--surface` Text in cards & decks	Aa 19.90:1 ✓ AA-L ✓ AA ✓ AAA	Aa 17.00:1 ✓ AA-L ✓ AA ✓ AAA
`--muted-foreground` / `--background` Labels, captions	Aa 7.41:1 ✓ AA-L ✓ AA ✓ AAA	Aa 7.76:1 ✓ AA-L ✓ AA ✓ AAA
`--muted-foreground` / `--surface-2` Labels in wells / inputs	Aa 7.03:1 ✓ AA-L ✓ AA ✓ AAA	Aa 6.89:1 ✓ AA-L ✓ AA ✕ AAA
`--primary` / `--background` BPM readout, links	Aa 5.13:1 ✓ AA-L ✓ AA ✕ AAA	Aa 11.01:1 ✓ AA-L ✓ AA ✓ AAA
`--primary-foreground` / `--primary` Primary button label	Aa 5.36:1 ✓ AA-L ✓ AA ✕ AAA	Aa 11.01:1 ✓ AA-L ✓ AA ✓ AAA
`--success` / `--surface` Analyzed status	Aa 5.02:1 ✓ AA-L ✓ AA ✕ AAA	Aa 10.72:1 ✓ AA-L ✓ AA ✓ AAA
`--warning` / `--surface` CUE, pending status	Aa 5.02:1 ✓ AA-L ✓ AA ✕ AAA	Aa 11.19:1 ✓ AA-L ✓ AA ✓ AAA

PAIR	LIGHT	DARK
`--destructive` / `--surface` Error clipping	Aa 6.47:1	Aa 6.76:1

--border is intentionally low-contrast and only used for decorative separators. Interactive control outlines use `--muted-foreground/50` or the focus ring to meet 1.4.11.

WCAG COMPLIANCE RULES

1.4.3 AA	Text contrast ≥ 4.5:1 (≥ 3:1 for 18px bold / 24px+).	1.4.6 AAA	Body text targets ≥ 7:1 on canvas in both themes.
1.4.11 AA	Control boundaries & focus rings ≥ 3:1 against adjacent colour.	1.4.1 A	Colour is never the only signal — badges, pads and bands always carry a text label.
2.1.1 A	Every action, including transport and cue setting, is keyboard-operable.	2.4.7 AA	focus-visible ring: 2px --ring with 2px offset, never removed.
2.5.8 AA (2.2)	Touch targets ≥ 24px; booth controls standardise on 44px.	2.3.3 AAA	prefers-reduced-motion disables vinyl spin and waveform easing.

KEYBOARD MAP

Space	Play / pause the focused deck
Shift Space	CUE — return to cue point
1 – 8	Trigger hot cue pad 1–8
Shift 1-8	Set pad at playhead
← →	Nudge playhead ±1 beat (Alt: ±10ms)
+ -	Waveform zoom in / out

`Tab`	Move focus in DOM order; visible 2px ring
`Esc`	Close dialog / drawer, restore focus to trigger
`/`	Focus library search

ARIA CONTRACT

Icon-only button

`aria-label="Play deck 1"`

Icon gets aria-hidden="true".

Toggle chip

`aria-pressed="true|false"`

State announced, label stays constant.

Hot cue pad

`aria-label="Hot cue pad 1, Intro, 00:01.50"`

Full context, not just “PAD 1”.

Tempo / EQ knob

`role="slider" aria-valuemin aria-valuemax aria-valuenow aria-valuetext`

valuetext: “+2.5 dB”.

Waveform

`role="img" aria-label="Waveform, 5:00, playhead 0:01"`

Pair with keyboard nudge controls.

Deck status

`aria-live="polite"`

Announces load / analyze complete.

Tabs (Audit, Cues, Sets...)

`role="tablist" · role="tab" aria-selected · aria-controls`

Arrow keys move between tabs.

Dialog / drawer

`role="dialog" aria-modal="true" aria-labelledby`

Focus trapped, Esc closes.

§ 07 / 07

Scalability & architecture

The target frontend is a Next.js App Router app sitting in front of the existing Python core. The domain/application/infrastructure layers stay untouched; the vanilla JSX bundle is migrated route by route.

FRONTEND FOLDER STRUCTURE

TREE

```
beatwave-web/
├── app/                          # Next.js App Router — routes only
│   ├── (booth)/
│   │   ├── layout.tsx            # Deck shell, keyboard provider, transport bar
│   │   ├── cues/page.tsx         # Cue Editor  (was web/js/app.jsx › CUES tab)
│   │   ├── audit/page.tsx        # Library audit table
│   │   ├── sets/page.tsx         # Set builder / session graph
│   │   └── copilot/page.tsx      # AI copilot (was web/copilot.html)
│   ├── api/                      # Thin BFF ⟶ Python services (FastAPI/bottle)
│   │   └── tracks/[id]/route.ts
│   ├── layout.tsx                # Fonts, ThemeProvider, <html class="bg-
background">
│   └── globals.css               # Tokens (this document, § 02)
├── components/
│   ├── ui/                       # Primitives: button, badge, toggle-chip, input
│   ├── deck/                     # vinyl, transport, eq-knob, tempo-fader
│   ├── waveform/                 # canvas renderer, zoom-toolbar, cue-markers
│   └── library/                  # track-table, dropzone, recognition-panel
├── features/                     # Vertical slices: hooks + components + api
│   ├── cues/{use-cues.ts, cue-pad-grid.tsx, cues.api.ts}
│   └── analysis/{use-analysis.ts, analysis.api.ts}
├── lib/
│   ├── audio/                    # Web Audio graph, decoding, worker bridge
│   ├── stores/deck-store.ts      # Zustand — real-time playback state
│   └── cn.ts
└── workers/waveform.worker.ts    # Peak extraction off the main thread
```

BACKEND BOUNDARY

PYTHON

```
# Existing Python core — unchanged, exposed via HTTP
domain/          models.py               # Track, Cue, Set entities
application/     analysis_service.py     # Use-cases (deck, stream, taxonomy)
infrastructure/  precision_engine.py     # DSP, FLAC, Shazam, Spotify adapters
services/        dsp_engine.py · mood_intelligence_service.py · copilot_service.py
routers/         audit · itunes · spotify · superflac · taxonomy
```

Next.js never imports Python. All calls go through `app/api/*` route handlers, which add the session token header used by the remote/tablet mode.

STATE MANAGEMENT PATTERNS

Server state → SWR

WHEN · Library, track metadata, analysis results, sets.

useSWR keyed by resource URL; mutate() optimistic updates after save; revalidateOnFocus off in the booth.

Real-time state → Zustand store

WHEN · Playhead, play/pause, tempo, EQ, active pad — 60 fps updates.

Subscribe with selectors; the waveform canvas reads via store.subscribe() outside React to avoid re-renders.

URL state → searchParams

WHEN · Active tab, selected track, zoom level, filters.

Shareable, back-button friendly; read in Server Components, write with router.replace.

Local UI state → useState

WHEN · Open drawers, hover previews, form drafts.

Kept in the leaf component. Lift only when two siblings need it.

BUILD SCALABILITY NOTES

Token-only styling

Components reference semantic tokens, never hex. A re-theme is a globals.css change, not a

Feature slices

Each feature owns its hooks, API calls and components. Cross-feature imports go

refactor.	through components/ui only.
Main-thread budget Decoding and peak extraction run in a Web Worker; the canvas renders from a transferable Float32Array.	**Route-level code splitting** Copilot and Sets load on demand; the Cue Editor bundle stays under 200 kB gzipped.
Offline-first booth Fonts and vendor assets are self-hosted (as today via build_offline.py) so the app runs without venue Wi-Fi.	**Contract tests** The BDD features (features/*.feature) stay the acceptance suite; the BFF is typed from the Python models.

HANDOVER CHECKLIST

Tokens merged into globals.css · primitives in components/ui with all five states · contrast matrix green for AA · keyboard map implemented · SWR + Zustand split respected · Lighthouse a11y ≥ 95 on Cue Editor.